

National Neighborhood Data Archive (NaNDA): Socioeconomic Status and Demographic Characteristics of ZIP Code Tabulation Areas, United States, 2008-2017



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Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains measures of socioeconomic and demographic characteristics by US ZIP code tabulation areas (ZCTAs) for the years 2008-2017. Example measures include population density; population distribution by race, ethnicity, age, and income; and proportion of population living below the poverty level, receiving public assistance, and female-headed families. The dataset also contains a set of index variables to represent neighborhood disadvantage and affluence.

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Data Sources

Population values and demographic proportions per ZCTA are from the American Community Survey five-year estimates for 2008-2012 (United States Census Bureau, 2012) and 2013-2017 (United States Census Bureau, 2017).

Land area for each ZCTA comes from the TIGER/Line shapefiles. The version released in 2019 (United States Census Bureau, 2019) was used to maintain consistency with other NaNDA datasets.

Coverage

The dataset contains one observation per ZCTA in the United States, excluding US island territories.

Methodology

This dataset contains key variables from the US Census Bureau's American Community Survey that describe the socioeconomic status and demographics of each census tract. It also contains three theoretically derived measures of neighborhood disadvantage, affluence and ethnic/immigrant concentration. Similar datasets at the census tract level have been created and published previously for 2000-2010 (Clarke & Melendez, 2019) and 2008-2017 (Melendez et al., 2020).

To construct this dataset, we extracted key census indicators related to race, ethnicity, age, income level, employment, poverty, and home ownership from the ACS 2012 five-year estimate (covering 2008-2012). We merged the variables with the same variables from the ACS 2017 five-year estimate (covering 2013-2017) and with each ZCTA's land area from the 2010 TIGER/Line shapefiles for ZIP code tabulation areas. We then used those variables to construct three indices as described below: neighborhood disadvantage, neighborhood affluence, and ethnic immigrant concentration.

Construction of the index variables was informed by previous work to construct a set of variables that would characterize the sociodemographic structure of census tracts over time. In 2000, we conducted a principal factor analysis with an orthogonal varimax rotation of ten census indicators (log transformed to correct positive skew) (Morenoff et al., 2007). Our aim was to derive a parsimonious set of factors that capture the shared variance of a broad spectrum of neighborhood structural characteristics. Results from the factor analysis indicated three separate factors:

- The first factor, which we interpret as neighborhood disadvantage, is characterized by high levels of poverty, unemployment, female-headed families, households receiving public assistance income, and a high proportion of African Americans in a census tract.
- The second factor represents a mix of characteristics associated with neighborhood affluence (concentrations of adults with a college education; with incomes > \$75K; and employed in managerial and professional occupations). Distinguished from other non-disadvantaged census tracts by their large share of high income, highly-educated, adults in professional occupations, affluent census tracts are likely to attract a set of institutions (e.g., food stores, places to exercise, well-maintained buildings and parks) that foster a set of norms (e.g., an emphasis on exercise and healthy diets) conducive to good health (Clarke et al., 2014). Distinct from simply being the absence of neighborhood disadvantage, neighborhood affluence is associated with higher levels of social control and leverage over local institutions that can foster social environments that facilitate health (Browning & Cagney, 2003).
- The third factor represents ethnic and immigrant concentration. Higher values indicate more Hispanic and foreign born in the census tract.

Results from this factor analysis were used to guide the creation of the neighborhood SES indices, which have been translated to the ZCTA level in this dataset.

- Neighborhood socioeconomic disadvantage is an average of five census indicators (proportion non-Hispanic Black, proportion of female headed families with children, proportion of households with public assistance income or food stamps; proportion of families with income below the federal poverty level; proportion of population age 16+ unemployed) ranging from 0 to 1.0.
 - Note: A separate neighborhood disadvantage index is created without non-Hispanic Black, including only four census indicators (proportion of female headed families with children, proportion of households with public assistance income or food stamps; proportion of families with income below the federal poverty level; proportion of population age 16+ unemployed). This measure should be used in statistical models when individual race is also included as a covariate.
- Neighborhood affluence is an average of three census indicators (proportion of households with income greater than \$75K, proportion of population age 16+ employed in professional or managerial occupations; proportion of adults with Bachelor's Degree or higher) ranging from 0 to 1.0.
- Neighborhood ethnic immigrant concentration is an average of two census indicators (proportion Hispanic, proportion foreign born).

Usage Note

Users should be aware that ZCTAs are not equivalent to ZIP codes. ZIP codes are linear mail delivery routes created by the US Postal Service. ZIP code tabulation areas are spatial features consisting of census blocks grouped by the predominant ZIP code found on the block (United States Census Bureau, 2020).

In some cases, a location's address is not the same as its ZCTA. For example, some ZIP codes represent single-point addresses such as large post offices or office buildings. Also, the ZIP code for an address may not match its ZCTA if the ZIP code is not the most common ZIP code on the block. See the Census Bureau's ZCTA overview at <https://www.census.gov/programs-surveys/geography/guidance/geo-areas/zctas.html> (United States Census Bureau, 2020) for more information on how ZCTA boundaries are calculated.

Users wanting to combine this dataset with ZIP code geocoded data must use a ZIP code to ZCTA crosswalk. Such a crosswalk is available on the UDS Mapper website at <https://www.udsmapper.org/zcta-crosswalk.cfm> (John Snow Inc., 2018). Sample code for merging the UDS Mapper crosswalk with NaNDA datasets is available on the NaNDA repository at <https://www.openicpsr.org/openicpsr/project/120088/> (Chenoweth & Khan, 2020).

Variables

Variable	Type	Obs	Unique	Mean	Min	Max	Label
zcta10	string	32,989	32,989	.	.	.	ZCTA code, 2010
aland10	float	32,989	32,975	86.8965	0.0019668	13467.65	ZCTA land area, square miles
totpop08_12	float	32,989	15,297	9370.474	0	115538	Total population, ACS 2008-2012
totpop13_17	float	32,989	15,466	9730.25	0	119204	Total population, ACS 2013-2017
popden08_12	float	32,989	32,650	1252.22	0	203546.2	Persons per square mile, ACS 2008-2012
popden13_17	float	32,989	32,669	1313.124	0	251590.5	Persons per square mile, ACS 2013-2017
phspanic08_12	float	32,653	24,189	0.0831359	0	1	Proportion of people of Hispanic origin, ACS 2008-2012
phspanic13_17	float	32,674	24,815	0.090881	0	1	Proportion of people of Hispanic origin, ACS 2013-2017
pnhwhite08_12	float	32,653	27,742	0.7872955	0	1	Proportion of people non-Hispanic White, ACS 2008-2012
pnhwhite13_17	float	32,674	27,962	0.7741433	0	1	Proportion of people non-Hispanic White, ACS 2013-2017
pnhblack08_12	float	32,653	20,810	0.0750392	0	1	Proportion of people non-Hispanic Black, ACS 2008-2012
pnhblack13_17	float	32,674	21,228	0.0752316	0	1	Proportion of people non-Hispanic Black, ACS 2013-2017
pfborn08_12	float	32,653	24,099	0.0583473	0	1	Proportion of people who are foreign born, ACS 2008-2012
pfborn13_17	float	32,674	24,318	0.0604658	0	1	Proportion of people who are foreign born, ACS 2013-2017
ped1_08_12	float	32,616	27,638	0.1469904	0	1	Proportion with Less than High School Diploma, ACS 2008-2012
ped1_13_17	float	32,633	27,398	0.1277573	0	1	Proportion with Less than High School Diploma, ACS 2013-2017
ped2_08_12	float	32,616	28,493	0.6350505	0	1	Proportion with High School Diploma and/or Some College, ACS 2008-2012
ped2_13_17	float	32,633	28,418	0.6353307	0	1	Proportion with High School Diploma and/or Some College, ACS 2012-2013
ped3_08_12	float	32,616	27,849	0.2179591	0	1	Proportion with Bachelor's Degree or Higher, ACS 2008-2012

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
ped3_13_17	float	32,633	28,021	0.2369121	0	1	Proportion with Bachelor's Degree or Higher, ACS 2013-2017
pin1b_08_12	float	32,246	21,715	0.0801488	0	1	Proportion of families with Income less than 15K, ACS 2008-2012
pin1b_13_17	float	32,215	21,310	0.0726971	0	1	Proportion of families with Income less than 15K, ACS 2013-2017
pin2b_08_12	float	32,246	23,093	0.1440926	0	1	Proportion of families with Income 15-30K, ACS 2008-2012
pin2b_13_17	float	32,215	22,663	0.1244416	0	1	Proportion of families with Income 15-30K, ACS 2013-2017
pin3b_08_12	float	32,246	23,456	0.2017493	0	1	Proportion of families with Income 30-50K, ACS 2008-2012
pin3b_13_17	float	32,215	23,225	0.1868742	0	1	Proportion of families with Income 30-50K, ACS 2013-2017
pin4b_08_12	float	32,246	24,112	0.3549839	0	1	Proportion of families with Income 50-100K, ACS 2008-2012
pin4b_13_17	float	32,215	23,921	0.3495263	0	1	Proportion of families with Income 50-100K, ACS 2013-2017
pin5b_08_12	float	32,246	23,678	0.2190254	0	1	Proportion of families with Income greater than 100K, ACS 2008-2012
pin5b_13_17	float	32,215	24,060	0.2664608	0	1	Proportion of families with Income greater than 100K, ACS 2013-2017
pincgt75k08_12	float	32,246	24,743	0.3621547	0	1	Proportion of families with Income greater than 75K, ACS 2008-2017
pincgt75k13_17	float	32,215	24,733	0.4128814	0	1	Proportion of families with Income greater than 75K, ACS 2013-2018
pnmvar08_12	float	32,652	28,649	0.2683473	0	1	Proportion of People 15+ Never Married, ACS 2008-2012
pnmvar13_17	float	32,673	28,748	0.278832	0	1	Proportion of People 15+ Never Married, ACS 2013-2017
p18yr_08_12	float	32,653	29,019	0.2221844	0	1	Proportion of population under 18 yrs, ACS 2008-2012
p18yr_13_17	float	32,674	28,912	0.2120259	0	1	Proportion of population under 18 yrs, ACS 2013-2017
p18_2908_12	float	32,653	28,404	0.1446233	0	1	Proportion of population 18-29 yrs, ACS 2008-2012
p18_2913_17	float	32,674	28,382	0.1450826	0	1	Proportion of population 18-29 yrs, ACS 2013-2017
p30_3908_12	float	32,653	27,966	0.1137566	0	1	Proportion of population 30-39 yrs, ACS 2008-2012
p30_3913_17	float	32,674	27,832	0.1135624	0	1	Proportion of population 30-39 yrs, ACS 2013-2017

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p40_4908_12	float	32,653	28,334	0.1390777	0	1	Proportion of population 40-49 yrs, ACS 2008-2012
p40_4913_17	float	32,674	28,058	0.1217779	0	1	Proportion of population 40-49 yrs, ACS 2013-2017
p50_6908_12	float	32,653	29,435	0.2690104	0	1	Proportion of population 50-69 yrs, ACS 2008-2012
p50_6913_17	float	32,674	29,299	0.2844066	0	1	Proportion of population 50-69 yrs, ACS 2013-2017
pge7008_12	float	32,653	28,385	0.1113475	0	1	Proportion of population 70+ yrs, ACS 2008-2012
pge7013_17	float	32,674	28,523	0.1231446	0	1	Proportion of population 70+ yrs, ACS 2013-2017
punemp08_12	float	32,652	25,988	0.0523445	0	1	Proportion 16+ civ labor force unemployed, ACS 2008-2012
punemp13_17	float	32,673	25,170	0.0371954	0	1	Proportion 16+ civ labor force unemployed, ACS 2013-2017
pprof08_12	float	32,413	26,645	0.3188083	0	1	Proportion emplyd civ 16+ mgmt/bus/sci/arts, ACS 2008-2012
pprof13_17	float	32,421	26,410	0.3302346	0	1	Proportion emplyd civ 16+ mgmt/bus/sci/arts, ACS 2013-2017
ppov08_12	float	32,464	28,694	0.1487878	0	1	Proportion people w/ income past 12 months below poverty level, ACS 2008-2012
ppov13_17	float	32,470	28,616	0.1469595	0	1	Proportion people w/ income past 12 months below poverty level, ACS 2013-2017
ppubas08_12	float	32,403	24,731	0.1214181	0	1	Proportion of households with public assistance income, ACS 2008-2012
ppubas13_17	float	32,420	24,822	0.131288	0	1	Proportion of households with public assistance income, ACS 2013-2017
pfhfam08_12	float	32,246	21,468	0.0874113	0	1	Proportion female-headed families w/ kids, ACS 2008-2012
pfhfam13_17	float	32,215	21,235	0.0821586	0	1	Proportion female-headed families w/ kids, ACS 2013-2017
pownoc08_12	float	32,403	25,704	0.7401848	0	1	Proportion owner occupied hus, ACS 2008-2012
pownoc13_17	float	32,420	25,655	0.7307068	0	1	Proportion owner occupied hus, ACS 2013-2017
disadvantage08_12	float	32,653	31,720	0.0971852	0	0.8484849	mean of pnhblack pfhfam ppubas ppov punemp, ACS 2008-2012
disadvantage13_17	float	32,674	31,670	0.0946882	0	0.8	mean of pnhblack pfhfam ppubas ppov punemp, ACS 2013-2017
disadvantage2_08_12	float	32,652	31,555	0.1024579	0	0.8901898	mean of pfhfam ppubas ppov punemp, ACS 2008-2012
disadvantage2_13_17	float	32,673	31,510	0.0992612	0	1	mean of pfhfam ppubas ppov punemp, ACS 2013-2017

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
affluence08_12	float	32,648	31,903	0.2986112	0	1	mean of pincgt75k ped3 pprof, ACS 2008-2012
affluence13_17	float	32,668	31,979	0.3255658	0	1	mean of pincgt75k ped3 pprof, ACS 2013-2017
ethnicimmigrant08_12	float	32,653	26,388	0.0707416	0	0.8808411	mean of phispanic pfborn, ACS 2008-2012
ethnicimmigrant13_17	float	32,674	26,519	0.0756734	0	1	mean of phispanic pfborn, ACS 2013-2017

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